

6-2-1: Examining the 802.3 Standard

At the end of this episode, I will be able to:

1. Compare and contrast various 802.3 Ethernet standards, given a scenario.

Learner Objective: *Compare and contrast various 802.3 Ethernet standards, given a scenario.*

Description: In this episode, the learner will examine the IEEE's local area network or LAN standards known as 802.3. We will explore the characteristics of these standards, such as the BASE naming convention and Speeds.

- 802.3a (Ethernet):
 - o Speed: 10 Mbps
 - o BASE standard: 10BASE2
 - o Transmission media: coaxial
- 802.3i (Ethernet):
 - o Speed: 10 Mbps
 - o BASE standard: 10BASE-T
 - o Transmission media: twisted-pair
- 802.3j (Ethernet):
 - o Speed: 10 Mbps
 - o BASE standard: 10BASE-F
 - o Transmission media: fiber optic
- 802.3u (Fast Ethernet):
 - o Speed: 100 Mbps
 - o BASE standard: 100BASE-TX
 - o Transmission media: twisted-pair, fiber optic
- 802.3z (Gigabit Ethernet):
 - o Speed: 1 Gbps
 - o BASE standard: 1000BASE-X (family of 1000BASE standards)
 - o Transmission media: fiber optic, copper
- 802.3ab (Gigabit Ethernet):
 - o Speed: 1 Gbps
 - o BASE standard: 1000BASE-T
 - o Transmission media: twisted-pair cable
- 802.3ae (10 Gigabit Ethernet):
 - o Speed: 10 Gbps
 - o BASE standard: 10GBASE-SR/LR/ER
 - o Transmission media: fiber optic
- 802.3an (10GBASE-T):
 - o Speed: 10 Gbps
 - o BASE standard: 10 Gbps
 - o Transmission media: twisted-pair
- 802.3bm (100G/40G Ethernet):
 - o Speed: 40 & 100 Gbps
 - o BASE standard: 100GBASE-SR4/CR4
 - o Transmission media: fiber optic, twisted-pair

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- Additional Resources:

° If applicable